

Subhasri Vijay

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EDUCATION

JOHNS HOPKINS UNIVERSITY, Baltimore, Maryland

Master of Science in Engineering: Computer Science

Graduated May 2026

Bachelor of Science: Computer Science; Minors: Computational Medicine, Entrepreneurship and Management

Graduated May 2025

Relevant Coursework: Data Structures (Java), Computer System Fundamentals (C, C++), Introduction to Algorithms, Machine Learning: Deep Learning, Artificial Intelligence, Computational Genomics: Sequencing, Sketching and Indexing, Full Stack: JavaScript, Databases, Human Computer Interaction, Computer Innovation and Entrepreneurship, Computer Vision, Machine Intelligence, Computing the Transcriptome, Machine Translation

SPECIALIZED SKILLS

- Software Skills:** Python Programming, RDMS, Java Programming, C Programming, MATLAB, R, UX/UI Design, Type Script, React, Java Script, Git, Backend, Front-end, SQL, Linux, PyTorch, TensorFlow, HTML, Figma
- Special skills:** Design Thinking, Critical Thinking, Communication, Team Player, Problem Solving

PROJECTS AND EXPERIENCE

Cataract Research Assistant, Wilmer Eye Institute

January 2022-May 2026

- Created and analyzed six surgeon-specific ground-truth strategies to benchmark model performance on the question: "Can AI accurately identify the expert surgeon?"; evaluated using accuracy, sensitivity, specificity, and precision. Findings presented at AUPO 2024.
- Curated and preprocessed 30+ cataract surgery videos in collaboration with clinicians to build high-quality, structured datasets for training and evaluating deep learning models.
- Applied and evaluated deep learning architectures (encoders, decoders) and metrics (AUC, confusion matrix) for surgical skill assessment, contributing to clinically relevant advancements in AI-driven surgical evaluation.

Co-Founder, RescueReady

January 2025-Sept 2025

- Co-founded and led development of an AI-powered EMT training platform that simulates high-pressure emergency scenarios to enhance decision-making and situational awareness.
- Built a full-stack web application using React, TypeScript, and SQLite, integrating generative AI to generate realistic, protocol-aligned simulations in collaboration with EMTs and healthcare professionals.
- Achieved 3rd Place at HopStart General Ventures and won the Pava Center for Entrepreneurship Award, responsible for product vision, technical architecture, and user testing.

IMAZER, Self-driven

July 2024-July 2024

- Designed and implemented a Python-based image resizing application with a GUI, enabling intuitive user interaction and seamless batch processing across formats.
- Integrated key Python libraries (e.g., PIL, Tkinter), implemented robust error handling, and optimized performance for efficient and reliable image manipulation.

CAREATHON, Self-driven

July 2024-July 2024

- Developed an accessibility tool using Python with speech-to-text, text-to-speech, and live translation capabilities to assist users with speech and hearing impairments.
- Leveraged advanced Python libraries for speech and language processing, contributing to inclusive healthcare communication and enabling expressive, multi-language interaction.

POSTERS AND PUBLICATIONS

- Poster titled "Can AI accurately answer: Who is the expert surgeon?" at AUPO 2024.
- Poster titled "Surgical Skill Assessment of Cataract Videos" at ISBI 2025.
- Paper titled "A Vision Foundation Model for Cataract Surgery Using Joint-Embedding Predictive Architecture" at MIDL 2025.

TEACHING EXPERIENCE

Head Teaching Assistant, Data Structures: EN.601.226.

February 2024-May 2026

- Guided students in debugging complex programming tasks and simplified Data Structures concepts into accessible explanations, fostering better understanding, problem-solving, and technical communication skills.
- Conducted weekly office hours, providing hands-on assistance and promoting student engagement through practical learning experiences.
- Graded over 100 project submissions and 200 exam submissions while conducting 50+ office hours, ensuring fair grading and collaborating with faculty to create an inclusive and effective learning environment.

LEADERSHIP EXPERIENCE

National Liaison for JHU Hindu Student Council

April 2022-April 2024

- Represented JHU at the Leadership Summit 2022 in NYC, organizing and coordinating technical aspects for events, enhancing community engagement.

Member of Women in CS, ACM, SWE and Amazon's Women of the World

August 2021-May 2026

- Participated in career events, webinars, and coding circles to network, develop technical skills, and promote diversity and inclusion within the tech community.

President for Software Engineering Club (SWECL)

January 2025-August 2025

- Directed club activities to enhance member engagement and foster skill development in software engineering through workshops and guest speaker sessions, discussion on different aspects of software engineering and its applications.

HONORS AND AWARDS

- Awarded Joel Dean Award for Excellence in Teaching by Computer Science department 2026
- Dean's List for 8 semesters at Johns Hopkins University and graduated with General and Departmental Honors 2025
- Awarded the Masson Fellowship for 2024 by the Computer Science department for Research. 2025